

Demand Load Calculation for Single Family Dwelling with or without Secondary Suite/Coach House/Garden Suite

Permit No: _____

Address: _____

Date: _____

Please complete the appropriate table cells and check boxes accordingly.

2024 CE Code Rule		Dwelling Units		
		1 Main (Principal)	2 Secondary Suite	3 Coach House/ Garden Suite
Identifying the total living areas - Rule 8-110. 8-200 1) a):		m ²	m ²	m ²
i) A basic load of 5000 W for the first 90 m ² or less of living area; plus				
ii) 1000 W for each 90 m ² or portion thereof; plus				
iii) Any electric space-heating loads using the demand factors from Section 62. 100% of the first 10kW, plus 75% for any amount over 10 kW; plus any air-conditioning loads with demand factor of 100%, subject to interlocks provided as per Rule 8-106 3); plus Δ Electric boiler, duct heater, furnace: 100%;	Baseboard Heaters:			
	AC:			
	Heat Pump:			
	Other:			
iv) Any electric range load: 6000 W for a single range plus 40% of any amount by which the rating of the range exceeds 12 kW; plus				
v) Any electric tankless water heaters or electric water heaters for steamers, swimming pools, hot tubs, or spas with a demand factor of 100%; plus				
vi) Except as permitted by *Rule 8-106 11), any electric vehicle supply equipment loads with a demand factor of 100%; plus	100% EVSE:			
	*Deduct. EVSE loads:			
vii) A) 25% of the rating of each load with a rating in excess of 1500 W, if an electric range has been provided for, or vii) B) 100% of the combined load up to 6000 W, plus 25% of the combined load that exceeds 6000 W, if an electric range has not been provided for;	Dryer:			
	HWT:			
	Other:			
If the principal dwelling unit & secondary suite are separately metered, load calculation as per 8-200 2); if a single utility meter is provided for dwelling unit & suite, load calculation as per 8-200 1) for basic loads plus electrical loads of units as if there is only one dwelling unit.		W	← Add total unmetered load below to column 1	N/A
Total Calculated Loads		W	W	W
**Main Circuit Breaker – Amps		A	A	A
Type & size of service/feeder (AWG / kcmil Cu / Al)				
The calculated load for the consumer’s service supplying the dwelling units above must be based on Rule 8-200 2): Excluding any electric vehicle supply equipment loads, any electric space-heating loads and any air-conditioning loads, calculate the total load of all the dwelling units using the Rule 8-202 3) a) i) to v); plus, the requirements of Rule 8-202 3) b) to e). Note: any electrical vehicle supply equipment loads, if they are supplied from a panelboard installed in a dwelling unit, with a demand factor of 100%.				
100% of the unit having the heaviest load: Rule 8-202 3) a) i); excluding any EVSE loads, electric space-heating loads, air-conditioning loads. (unit 1 / 2)				
65% of the unit having the same or next smaller load: Rule 8-202 3) a) ii); excluding any EVSE loads, electric space-heating loads, air-conditioning loads. (unit 1 / 2)				

65% of the unit having the same or next smaller load: Rule 8-202 3) a) ii); excluding any EVSE loads, electric space-heating loads, air-conditioning loads (unit 2 / 3)							
Baseboard heating load,100% of the first 10 kW, 75% for any amount over 10 kW: Rule 8-202 3) b), Subject to Rule 8-106 3);							
Electric space-heating loads: Rule 8-202 3) b), Subject to Rule 8-106 3);							
Air-conditioning loads: Rule 8-202 3) c) iii), Subject to Rule 8-106 3);							
Heat Pump Motor Loads: Rule 8-202 3) c) iii), Subject to Rule 8-106 3);							
Where interlocks are installed, determine the larger ac / heating load from above: Rule 8-106 3): larger load to be considered in the determination of the calculated load.							
100% of EVSE loads supplied from units’ panel: ***As per Rule 8-202 1) a) vii);							
1) a) vii); Rule 8-202 3) d) and Rule 8-202 3) e). Not applicable				For loads not located in dwelling units			
Total Amps:		A		Total Watts:		W	
Ampacity Rating of the Service and Service Characteristics							
Service Ampacity:		A		Voltage:		V	
				Phase:		1 Ø 3W 3 Ø 4W	
Type & size of conductors: AWG/kcmil cu/Al		Overhead:		Underground:		Number of BC Hydro meters?	

Electrical Contractor Name & License Number	Email	Phone Number
FSR Name & Registration Number	Signature	Phone Number

Clear Form

Additional Requirements and Details
*Rule 8-200 1) a) vi) applies to the service or feeder supplying a single dwelling only. See Rule 8-200 2). **Precise rating or next higher standard rating of O/C must be used, if not, size the service for the anticipated ampere rating of O/C devices. ***Despite Rule 8-106 11), EVSE loads described in Rule 8-202 1) a) vii) are required to be excluded in accordance with Rule 8-200 2) a). Δ Other heating loads includes: heating cable sets, heating panel sets, heat pump auxiliary element, Patio heaters, Duct heaters, Electric furnace. Provide actual air conditioner or heat pump motor loads: Calculate NAMEPLATES FLA x Volts = Watts BC Hydro may restrict 400A services in some areas. When 2 supply services (2 drops) are provided by BC Hydro, submit separate load calculations with compliance of Rule 8-200 1) a) or b). Where EVSE loads are controlled by an EVEMS, the demand load for the EVSE shall be equal to the maximum load allowed by the EVEMS. i.e., If an EVEMS has a fixed rating of 50A is installed to Control 4 EVSE loads, then 50A must be used in EVSE load calculations in accordance with Section 8 of the CE Code. Where an Electric Vehicle Energy Management System (EVEMS) is used, the EVEMS must meet all the applicable conditions of CE Code Rule 8-106 11) and it must control the demand load for the EVSE in accordance with CE Code Rule 8-500. So far, a dedicated certification / product standard does not exist for the electrical equipment comprising EVEMS. "Approved" is defined by the CE Code. EVEMS equipment could be made approved by a Special Inspection to the CSA Model Field Evaluation Code, SPE 1000.